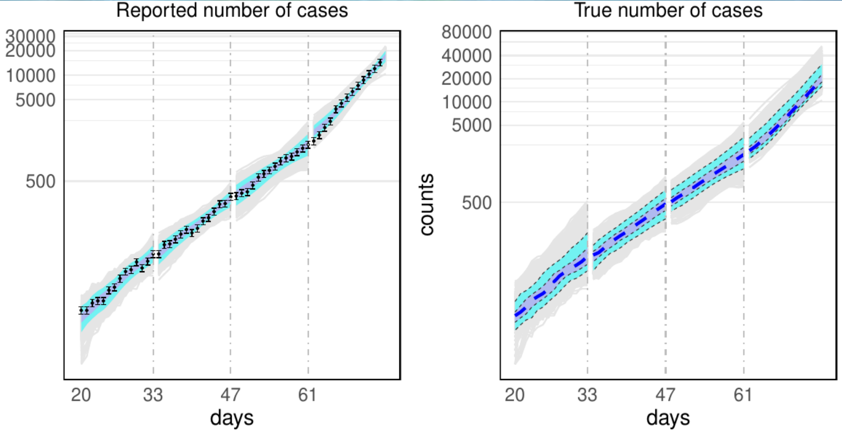
ParSocial workshop
IPDPS 2024

MAY 31, 2024

TOWARDS IMPROVED UNCERTAINTY QUANTIFICATION OF STOCHASTIC EPIDEMIC MODELS USING SEQUENTIAL MONTE CARLO



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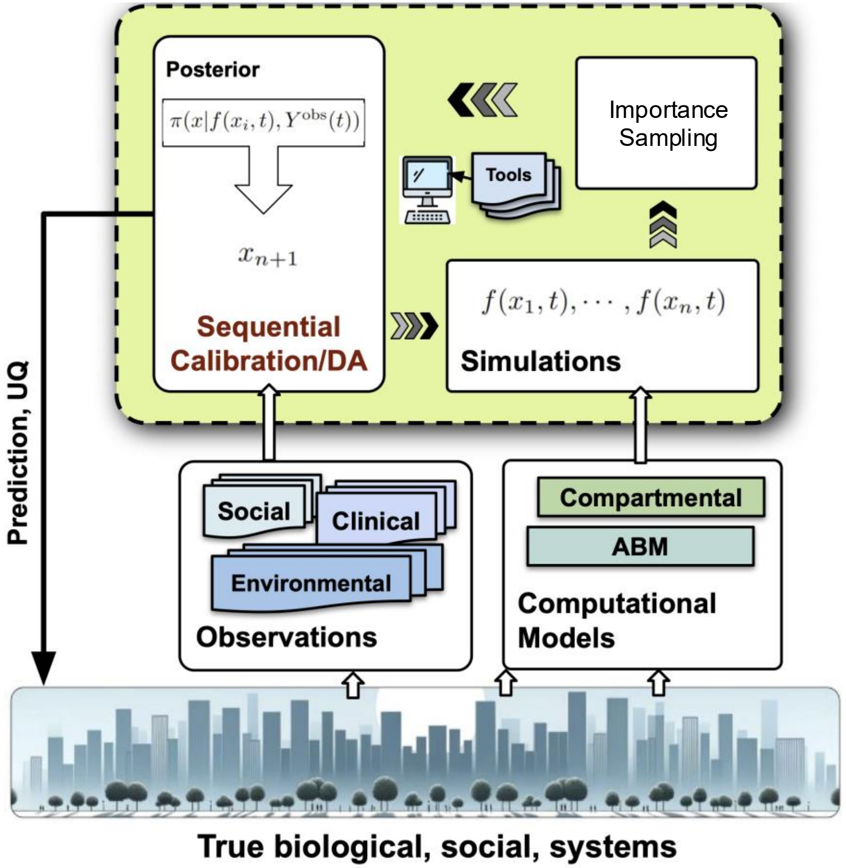
PREVIEW

- **Context**
  - — Epi-models connect *social and behavioral* dynamics to public health relevant outcomes.
  - — Tracking *time-varying* epidemic dynamics
  - — Data assimilation using *multiple sources of epi-data*
  - — *Uncertainty quantification* with multiple sources of uncertainty
  - — *Automated* workflows using HPC resources

- **Approach**
  - — Trajectory-oriented calibration framework
  - — Sequential Monte-Carlo for parameter inference

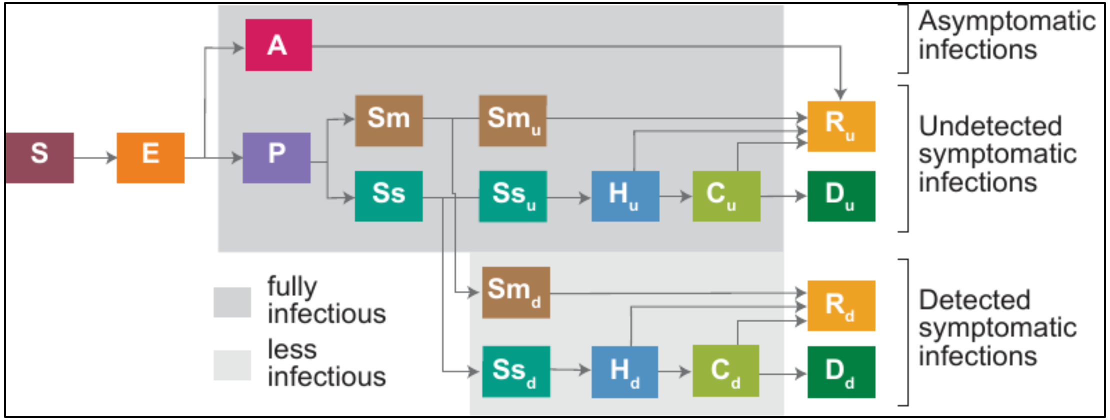


A STOCHASTIC COMPARTMENTAL MODEL TO DEMONSTRATE OUR APPROACH

- Stochastic compartmental model with disease specific states

- **Input**: time-dependent disease transmission rate
- **Output**: number of hospitalizations and deaths

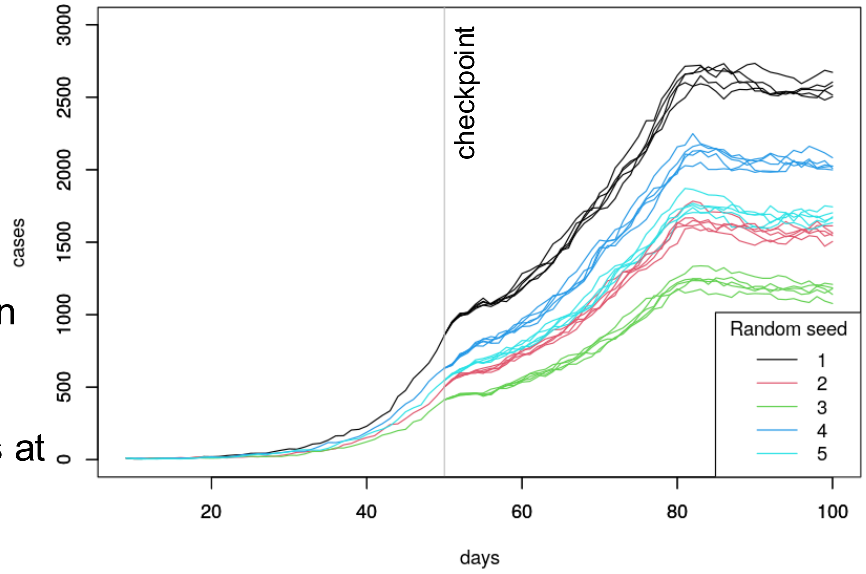
- Developed by the malaria modeling team at Northwestern University https://www.numalariamodeling.org/
    - Argonne (us) and Northwestern were two of the four modeling groups in the Illinois Governor's COVID-19 task force to support CDPH and IDPH



Runge, M., Richardson, R. A., Clay, P. A., Bell, A., Holden, T. M., Singam, M., ... & Gerardin, J. (2022). Modeling robust COVID-19 intensive care unit occupancy thresholds for imposing mitigation to prevent exceeding capacities. PLOS global public health, 2(5), e0000308.

COVID-19 SIMULATOR

- A stochastic model returns a different output when run repeatedly at the same input parameter setting.
- Random seeds are used as additional inputs and govern the sequence of draws from a (pseudo) random number generator.
  - Each simulated trajectory (combination of parameter input and random seed) defines a specific history with probabilistic infections, disease progression, and population mixing in the heterogeneous contact network.
- Checkpointing is used to serialize simulation states at specified times and then restart them from those times to continue forward.



CALIBRATION OBJECTIVES AND CHALLENGES

- The model parameters need to be estimated from the empirical data.

- **Challenges**:
  - Stochastic simulations.
  - Parameters can vary over time (e.g., disease transmissibility).
  - Some reported data are biased (e.g., under-ascertainment (reporting), delays in reporting).
  - Model needs to be readjusted often to reflect the changing dynamics (e.g., new variant)
  - The simulation history needs to be preserved to track unique population mixing into the future.

- **Proposed solution**:
  - Use random seed as a calibration input.
  - A statistical framework to use multiple data sources in calibration and to model the bias in the empirical data.
  - Sequential inference of the model parameters and uncertainty via Sequential Monte-Carlo.

APPROACHES WE DEVELOPED

Bias model

- The empirical data may contain biases (e.g., under-reporting of case counts, lag, etc.) which can affect model parameter estimation.
- We propose a statistical "measurement" model to track and estimate the proportion at which the reported case-counts are affected.

Sequential estimation

- The model calibration task is divided into several data-assimilation steps that are executed sequentially in time.
- In each data-assimilation window, importance sampling is used to estimate the model parameters.
- This sequential strategy along with the measurement bias model allows for:
  - unbiased estimation of time-varying disease parameters within a simple calibration framework
  - frequent model realignment with empirical data to track the changing dynamics
  - shorter turn-around time to updated model forecasts to support policy makers

CONNECTING MODELS TO DATA

Linking data to model

Random seed

$$\underbrace{\boldsymbol{y}_{1:T}}_{\text{observation}} = \underbrace{\boldsymbol{\eta}_{1:T}(\boldsymbol{\theta}, \boxed{s})}_{\text{simulation}} + \underbrace{\boldsymbol{\varepsilon}_T}_{\text{white noise}}, \quad \boldsymbol{\varepsilon}_T \sim MVN(0, \Sigma_T^2)$$

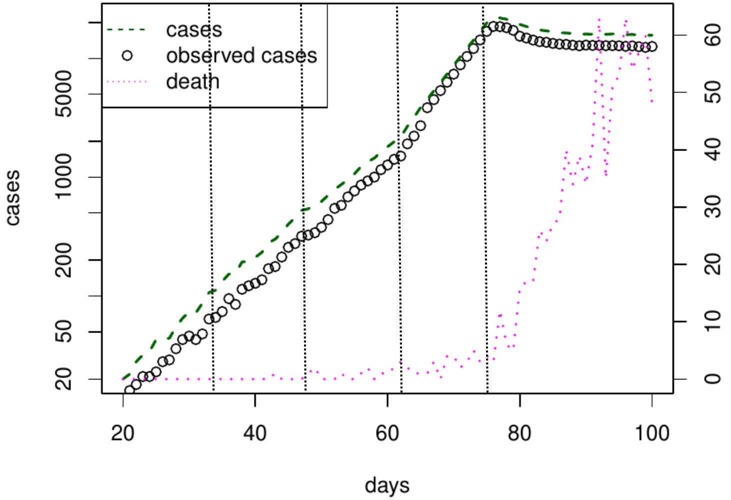
Incorporating data bias

$$\boldsymbol{y}_t = \boldsymbol{\eta}_t^{\text{obs}}(\boldsymbol{\theta}, s, \rho_t) + \boldsymbol{\varepsilon}_t, \quad \varepsilon_t \sim N(0, \sigma_t^2), 0 < \rho_t < 1$$

$$\underbrace{\boldsymbol{\eta}_t^{\text{obs}}(\boldsymbol{\theta}, s, \rho_t)}_{\text{Simulated reported counts}} \sim \text{Binomial}(\boldsymbol{\eta}_t(\boldsymbol{\theta}, s), \boxed{\rho_t})$$

Simulated reported counts

Additional parameter



A simulated ground-truth using SEIR COVID model which simulates COVID cases and deaths. The observation/bias model simulates the reported counts which are observed in reality instead of true case counts. The vertical dashed lines are time windows in which inference is performed.

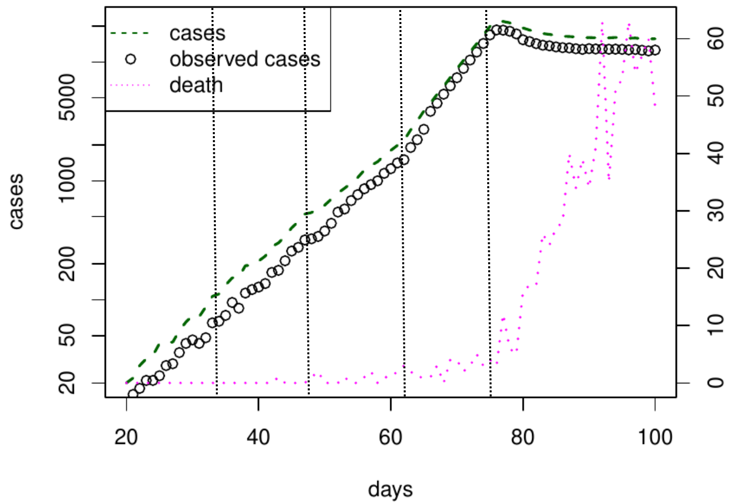
CONNECTING MODELS TO DATA

Likelihood

$$l(\boldsymbol{y}_{1:T}|\boldsymbol{\theta}, s, \rho_t) = \frac{1}{\sqrt{2\pi}|\Sigma_T|^{1/2}} \times$$
$$\exp\left\{ -\frac{1}{2}\left(\boldsymbol{y}_{1:T} - \boldsymbol{\eta}^{\text{obs}}_{1:T}(\boldsymbol{\theta}, s, \rho_t)^T\right) \Sigma_T^{-1}\left(\boldsymbol{y}_{1:T} - \boldsymbol{\eta}^{\text{obs}}_{1:T}(\boldsymbol{\theta}, s, \rho_t)\right)\right\}$$

Posterior

$$\begin{aligned}
&\pi\left(\boldsymbol{\theta}, s, \rho|\boldsymbol{y}^c_{1:T}, \boldsymbol{y}^d_{1:T}\right) \\
&\propto\ l\left(\boldsymbol{y}^c_{1:T}|\boldsymbol{\theta}, s, \rho\right) \times l\left(\boldsymbol{y}^d_{1:T}|\boldsymbol{\theta}, s, \rho\right) \times \pi(\boldsymbol{\theta}, s, \rho) \\
&\propto\ \text{MVN}\left(\boldsymbol{y}^c_{1:T}|\boldsymbol{\eta}^c_{1:T}(\boldsymbol{\theta}, s, \rho), \Sigma^c_t\right) \times \\
&\quad \text{MVN}\left(\boldsymbol{y}^d_{1:T}|\boldsymbol{\eta}^d_{1:T}(\boldsymbol{\theta}, s, \rho), \Sigma^d_t\right) \times \\
&\quad \pi(\boldsymbol{\theta}) \times \pi(s) \times \pi(\rho)
\end{aligned}$$



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CONNECTING MODELS TO DATA

Sequential posterior

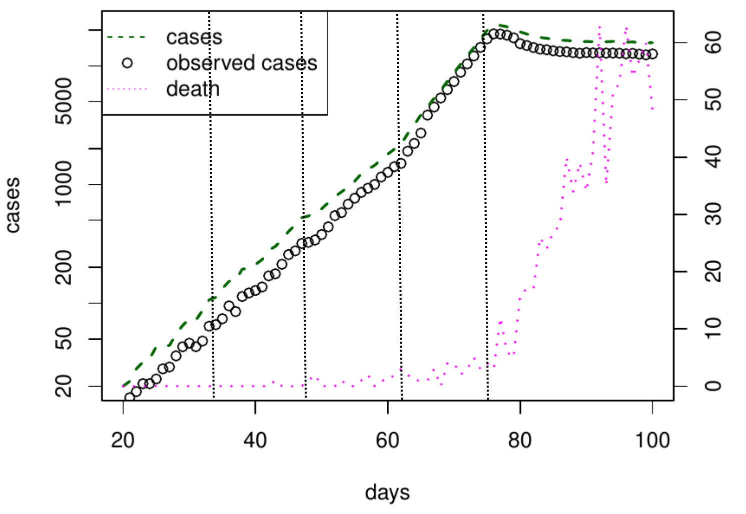
$$\pi\left(\boldsymbol{\theta}, s, \rho \middle| \boldsymbol{y}^c_{1:t_2}, \boldsymbol{y}^d_{1:t_2}\right) = \frac{l\left(\boldsymbol{y}^c_{1:t_2}, \boldsymbol{y}^d_{1:t_2} \middle| \boldsymbol{\theta}, s, \rho\right)\pi\left(\boldsymbol{\theta}, s, \rho\right)}{\pi\left(\boldsymbol{y}^c_{1:t_2}, \boldsymbol{y}^d_{1:t_2}\right)}$$

$$= \frac{l\left(\boldsymbol{y}^c_{1:t_1}, \boldsymbol{y}^d_{1:t_1} \middle| \boldsymbol{\theta}, s, \rho\right)\pi\left(\boldsymbol{\theta}, s, \rho\right)}{\pi\left(\boldsymbol{y}^c_{1:t_1}, \boldsymbol{y}^d_{1:t_1}\right)} \times \frac{l\left(\boldsymbol{y}^c_{(t_1+1):t_2}, \boldsymbol{y}^d_{(t_1+1):t_2} \middle| \boldsymbol{y}^c_{1:t_1}, \boldsymbol{y}^d_{1:t_1}, \boldsymbol{\theta}, s, \rho\right)}{\pi\left(\boldsymbol{y}^c_{(t_1+1):t_2}, \boldsymbol{y}^d_{(t_1+1):t_2} \middle| \boldsymbol{y}^c_{1:t_1}, \boldsymbol{y}^d_{1:t_1}\right)}$$

$$\boxed{\propto \pi\left(\boldsymbol{\theta}, s, \rho \middle| \boldsymbol{y}^c_{1:t_1}, \boldsymbol{y}^d_{1:t_1}\right) \times l\left(\boldsymbol{y}^c_{(t_1+1):t_2}, \boldsymbol{y}^d_{(t_1+1):t_2} \middle| \boldsymbol{y}^c_{1:t_1}, \boldsymbol{y}^d_{1:t_1}, \boldsymbol{\theta}, s, \rho\right)}$$

Prior for the next stage

Likelihood of the new data



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CALIBRATING COVID MODEL FOR ONE TIME-WINDOW

Importance sampling

Algorithm 1 Pseudo-code for single window IS

Require: N (total simulation budget)

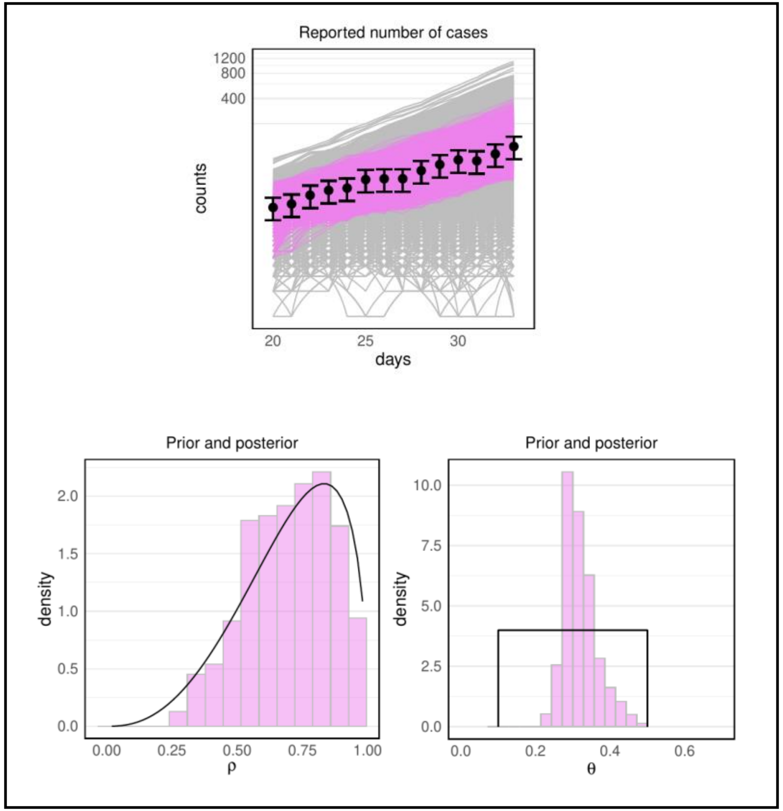
1: Sample: $(\boldsymbol{\theta}_i, s_i, \rho_i) \sim \pi(\boldsymbol{\theta}, s, \rho), \quad i = 1 : N.$

2: Run simulation: $\boldsymbol{\eta}^c_{1:t_1}(\boldsymbol{\theta}_i, s_i, \rho_i), \boldsymbol{\eta}^d_{1:t_1}(\boldsymbol{\theta}_i, s_i, \rho_i).$

3: Compute weight:

$$w_i^{(t_1)} \propto l\left(\boldsymbol{y}^c_{1:t_1} \middle| \boldsymbol{\theta}_i, s_i, \rho_i\right) \times l\left(\boldsymbol{y}^d_{1:t_1} \middle| \boldsymbol{\theta}_i, s_i, \rho_i\right)$$

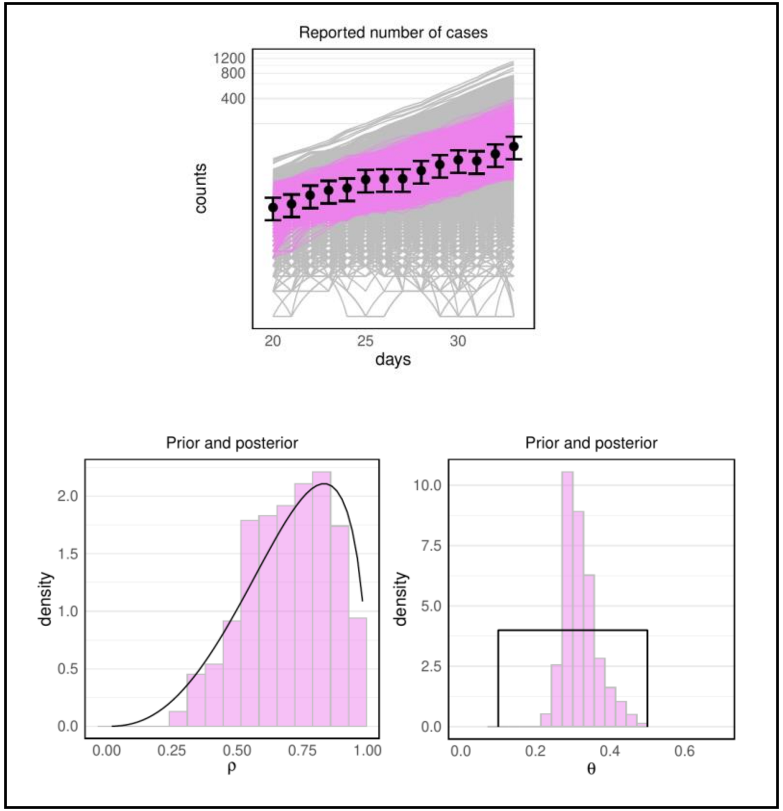
4: Resample: Sample $(\boldsymbol{\theta}_1, s_1, \rho_1), \ldots, (\boldsymbol{\theta}_N, s_N, \rho_N)$ with probabilities proportional to $w_i^{(t_1)}$.



Calibration using only reported case counts within the first time window -- (top) The simulated ground truth is shown by black dots which are used to find the posterior trajectories (in purple) that are consistent with the ground truth from the prior trajectories (in grey), (center) the prior density (in black) and posterior of the rho parameter in the binomial bias model, (right) the prior and posterior distribution of simulation input.

OPPORTUNITIES TO SPEED-UP THE FRAMEWORK

- Simulations are executed in parallel.
- Serializing the simulations that can be restored in future.
- Event-based automation (e.g., arrival of new data).



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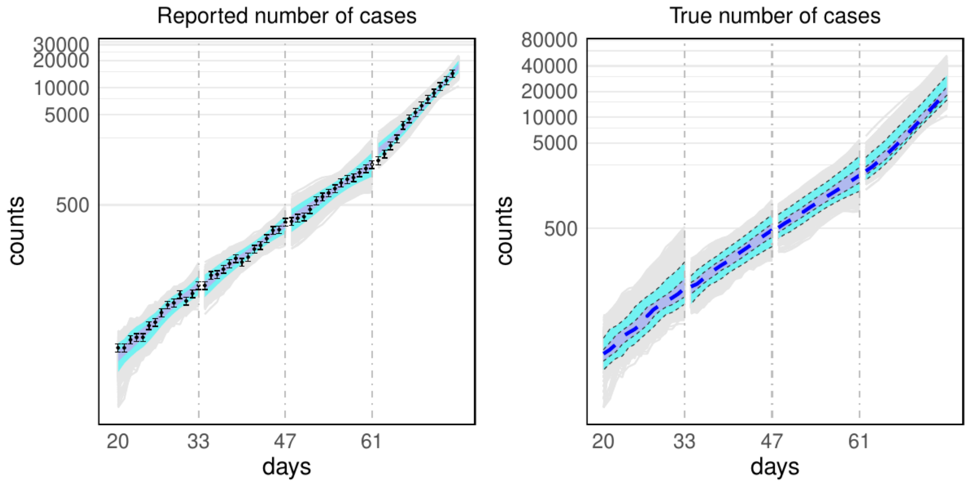
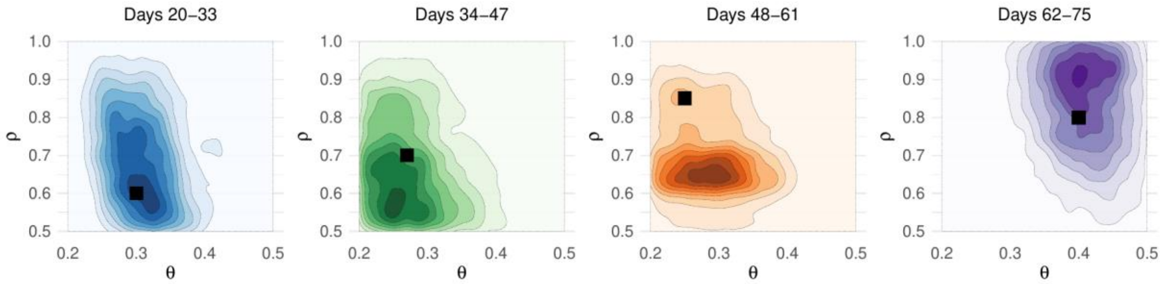
WHAT CAN WE INFER ABOUT TRANSMISSION AND REPORTING BIAS WHEN WE USE ONLY A SINGLE (REPORTED CASES) DATA SOURCE

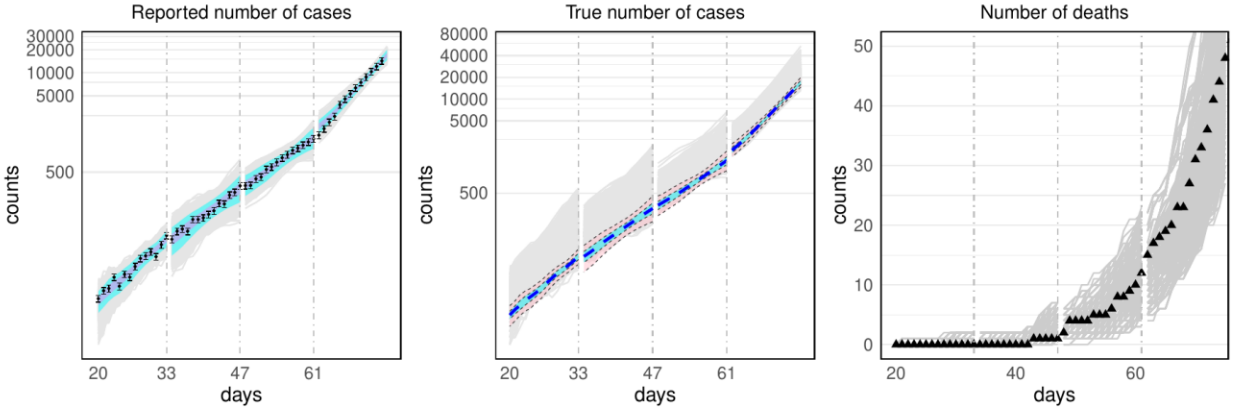




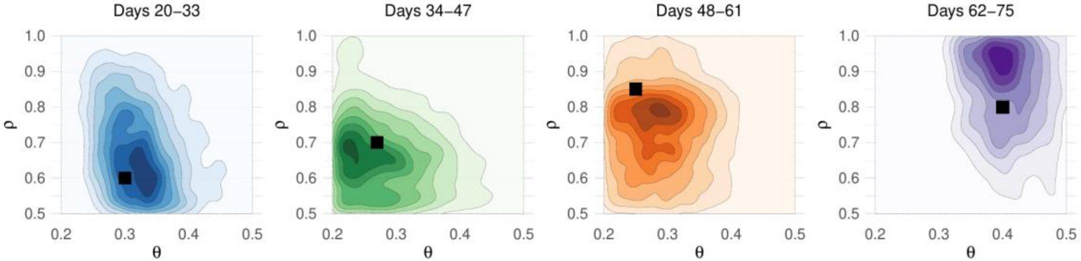
(top) Reported case counts (in black circles) are used to constrain the model parameters resulting in the posterior model output shown by grey lines. Purple and cyan shaded regions show 50% and 90% posterior credible intervals on the model output. The left plot shows the posterior on reported case counts and the right plot shows the posterior prediction on actual case counts that are unobserved. The blue dashed line represents the median predicted actual case counts. The vertical dashed grey lines represent the time boundaries for each calibration time window.

(bottom) 2d contour plot of the joint posterior of model parameters (theta, rho) for four time windows respectively for which sequential calibration is performed. Black square denotes the true value of the parameters using which the ground truth is generated.

WHAT CAN WE INFER ABOUT TRANSMISSION AND REPORTING BIAS WHEN WE USE BOTH REPORTED CASES AND DEATHS



(top) Reported case counts (in black circles) and deaths (in black triangles) are used to constrain the model parameters resulting in the posterior model output shown by grey lines. Purple and cyan shaded regions show 50% and 90% posterior credible intervals on the model output. From left to right, the plots show the posterior on reported case counts, actual case counts that are unobserved, and death counts respectively. The blue dashed line in the middle plot represents the median predicted actual case counts. The vertical dashed grey lines represent the time boundaries for each calibration time window.



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WHERE ARE WE NOW AND WHERE ARE WE GOING?

- Developed an approach for holistic incorporation of input parameter, measurement, and model uncertainties
  - Integrate many sources of uncertainty
  - Prediction uncertainty is important for decision making under uncertainty through robust decision making (RDM) approaches
- We track time-varying epidemic dynamics
  - Sequentially explore the disease trajectories with the help of simulation checkpointing and sequential inference
  - Imperfect models and data can be useful together
- Decision makers need operational approaches that incorporate the best knowledge “now” to assess the future
- These are use-inspired requirements that have guided the development of open science HPC

REQUIREMENTS FOR AN OPEN SCIENCE PLATFORM FOR ROBUST EPIDEMIC ANALYSIS

Developing Distributed High-performance Computing Capabilities of an Open Science Platform for Robust Epidemic Analysis

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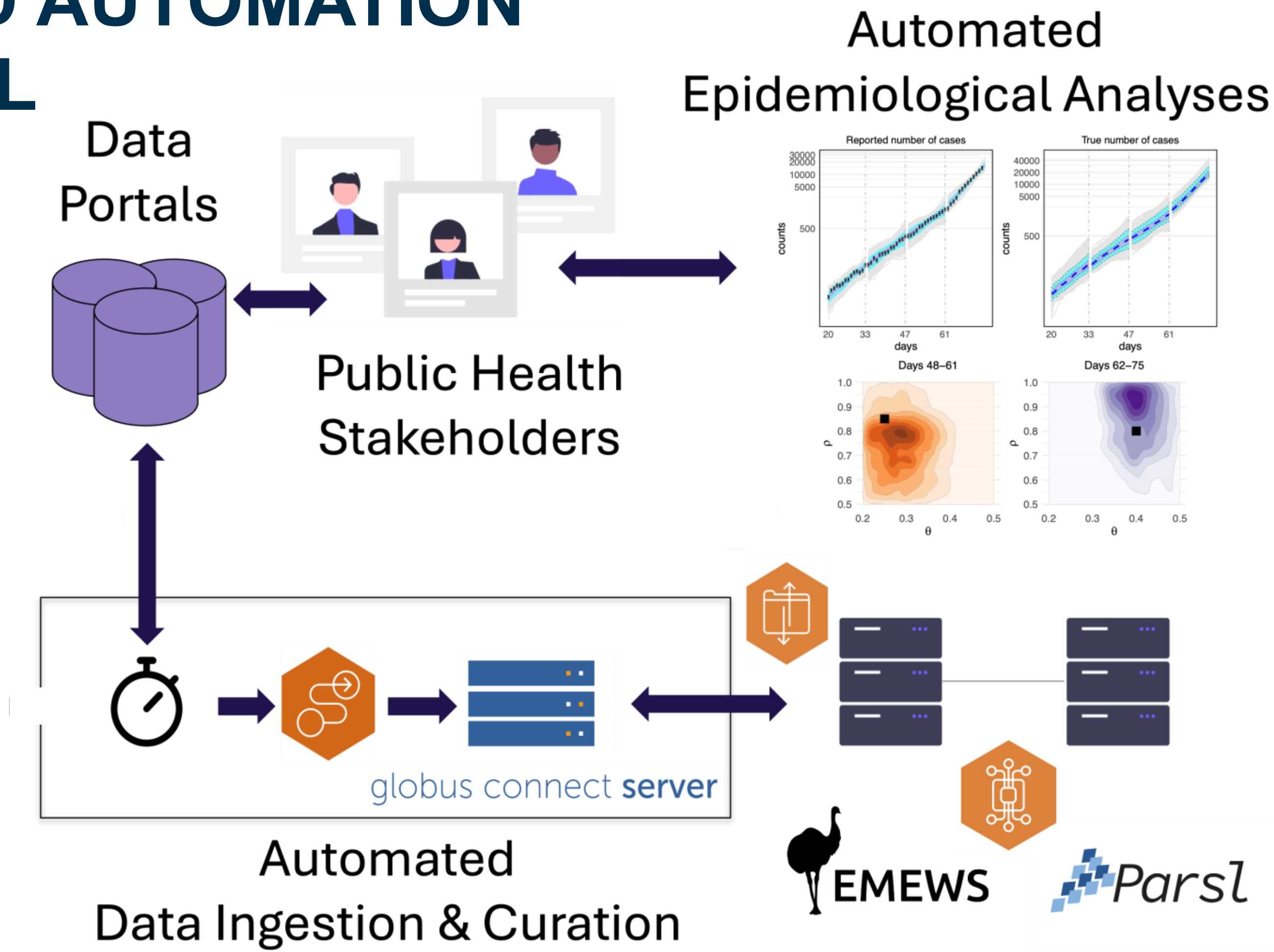
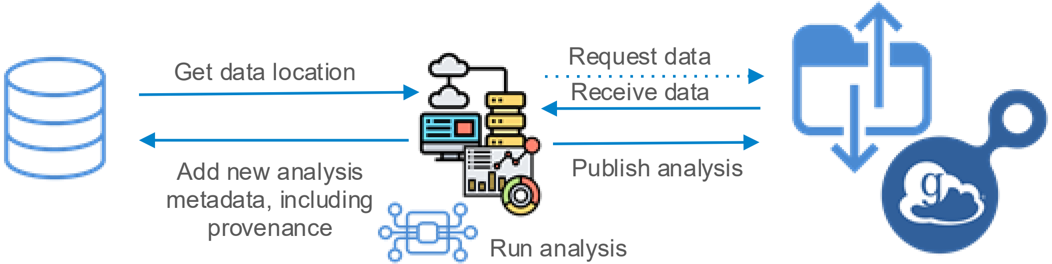
END TO END AUTOMATION IS THE GOAL



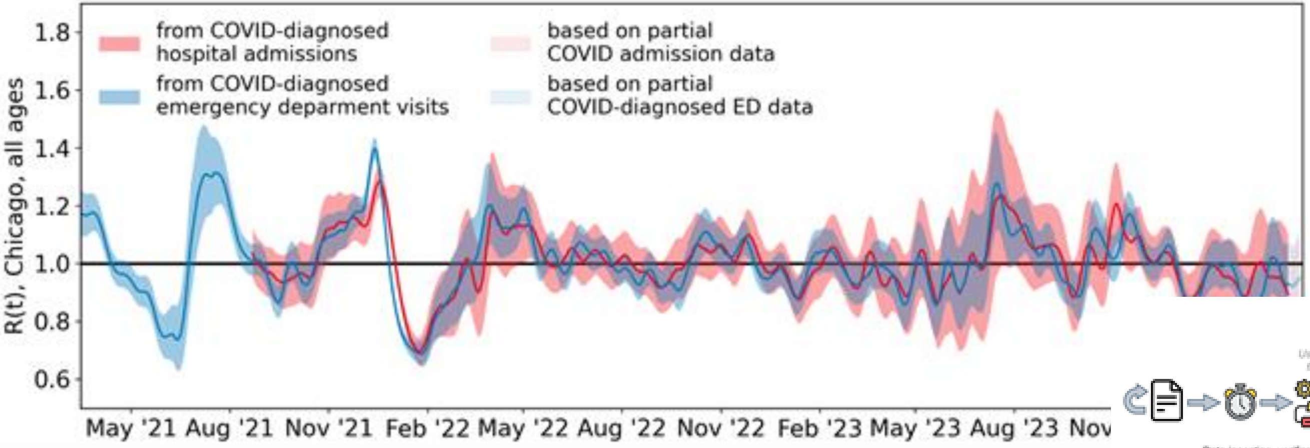
CLOUD SERVICES ENABLES "REAL-TIME" DATA ANALYSIS

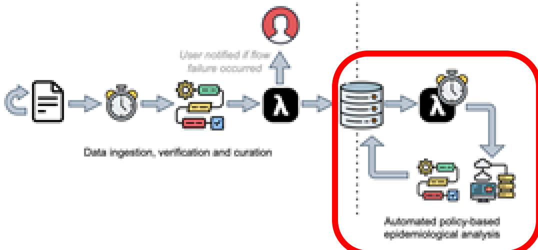
Event-based triggers enable analysis of data as new data is added

Data analysis flow:



Executed on: Polaris



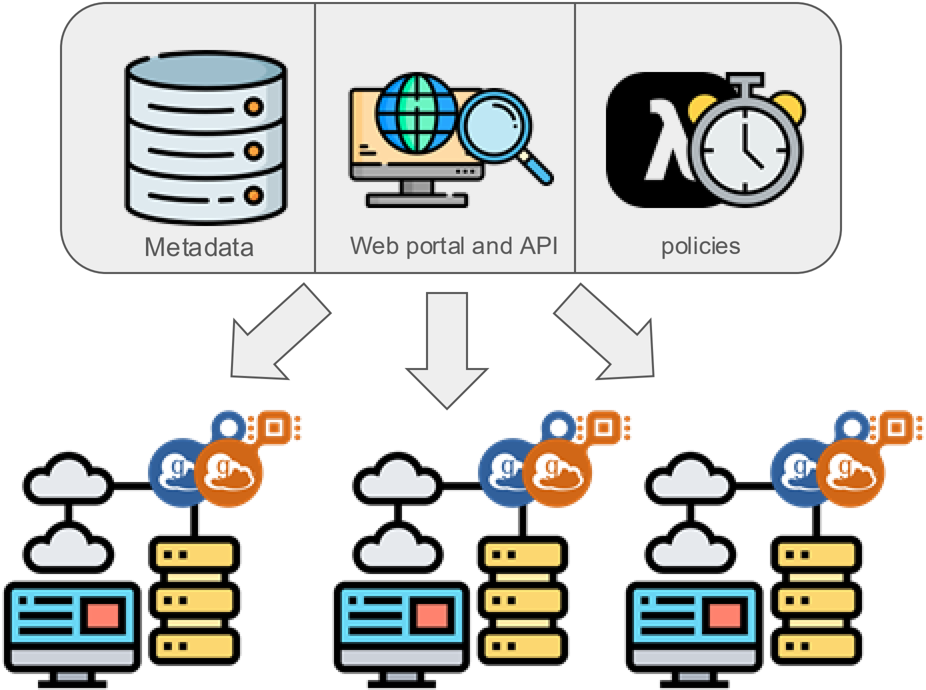


SCALING IS ACHIEVED VIA DECENTRALIZED COMPUTING AND STORAGE

Utilizing a hybrid model enables “bring-your-own resources” architecture, ensuring scalability as community of users grow

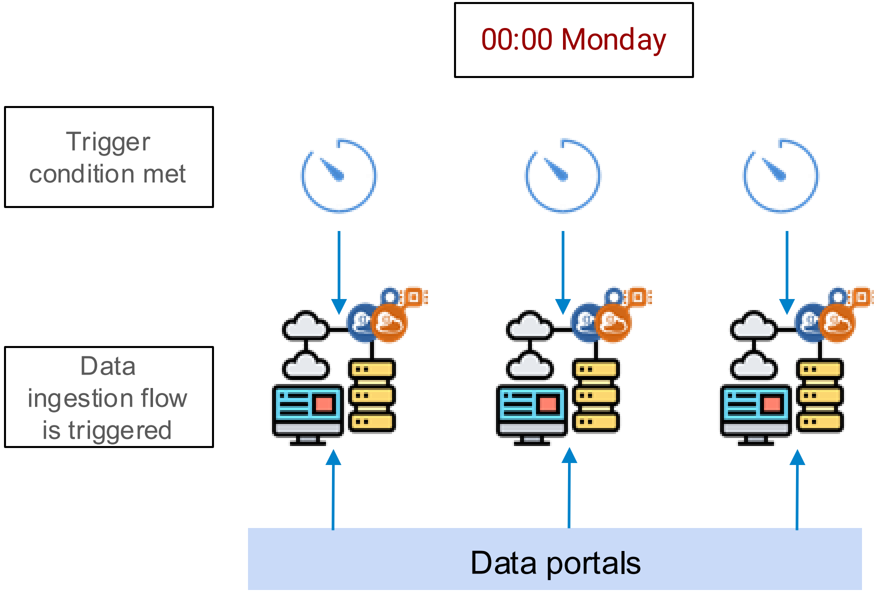
Internally managed metadata, search, and policies

Externally-managed compute and storage resources

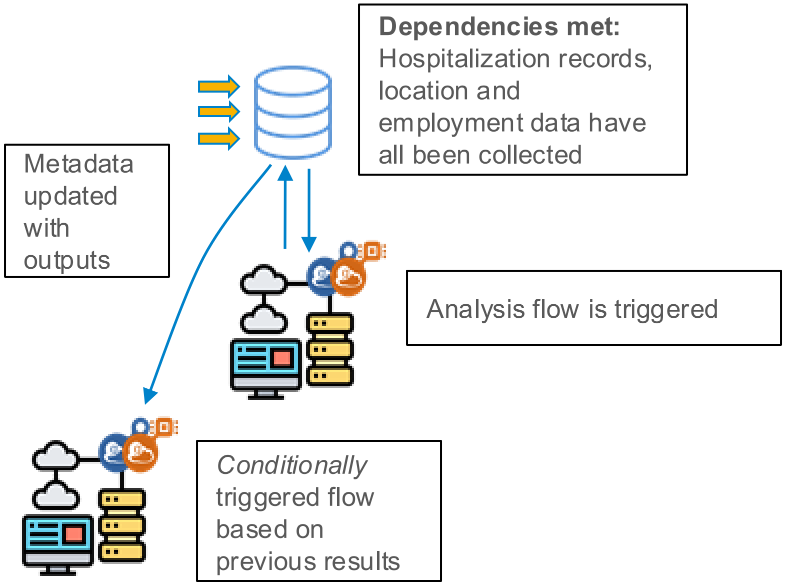


PERIODIC AND EVENT-BASED POLICIES ENABLE AUTONOMOUS SCIENCE

Periodic



Event-based



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- DOE ASCR BRaVE: Calibration Techniques and a High-Performance Computing Workflow for Disease Models

Questions?

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